



YEAR 12 Mathematics Ext 2

HSC Course

Assessment Task 3

2015

1. There are 3 sections
2. Multiple choice questions are 1 mark each. Marks allocated to each free response question are indicated.
3. Answer the multiple choice section on the response sheet provided.
4. Start each section on a new page. You are to hand in the integration and conics free response sections separately.
5. Calculators may be used
6. Time allowed - **60 minutes plus 5 minutes reading**

Topic	Mark
1. Integration section including Q1 and Q2 from multiple choice	/22
2. Integration section including Q3 and Q4 from multiple choice <i>Conics</i>	/20

TOTAL

/42

Multiple Choice Section (1 mark each)

Answer each question on the multiple choice answer sheet provided.

1. The definite integral

$$\int_{e^3}^{e^4} \frac{1}{x \ln x} dx$$

can be written in the form

$$\int_a^b \frac{1}{u} du$$

where

- (A) $u = \ln(x), a = \ln 3, b = \ln 4$
- (B) $u = \ln(x), a = 3, b = 4$
- (C) $u = \ln(x), a = e^3, b = e^4$
- (D) $u = \frac{1}{x}, a = e^3, b = e^4$

- 2) What is the value of

$$\int_0^1 \frac{e^x}{1 + e^x} dx$$

- (A) $\ln(1 + e)$
- (B) 1
- (C) $\ln\left(\frac{1+e}{2}\right)$
- (D) $\ln\left(\frac{e}{2}\right) - 2$

- 3) P is any point on the hyperbola with equation $x^2 - \frac{y^2}{4} = 1$.

If m is the gradient of the hyperbola at P , then m could be

- (A) *any real number x*
- (B) $-4 < x < 4$, where x is real
- (C) $-2 < x < 2$, where x is real
- (D) $x > 2$ or $x < -2$, where x is real
- 4) In the equation $Ax^2 + Cy^2 = F$ if $AC > 0$, $A < C$ and $F > 0$, then the curve is
- (A) an ellipse with a longer axis along the x axis
- (B) an ellipse with a longer axis along the y axis
- (C) a hyperbola intersecting the x axis
- (D) a hyperbola intersecting the y axis

(END OF MULTIPLE CHOICE QUESTIONS)

Integration Section (Start a new page)

MARKS

- 1) Find

$$\int \frac{x}{\sqrt{9 - 16x^2}} dx$$

2

- 2) Find

$$\int \frac{x^2}{x + 1} dx$$

2

- 3) Evaluate

$$\int_0^{\ln 3} x e^x dx$$

3

4) (a) Find real numbers A,B and C such that $\frac{2}{(x+1)(x^2+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$ 3

(b) Hence evaluate

$$\int_0^1 \frac{2}{(x+1)(x^2+1)} dx$$
 3

(c) By using the substitution $t = \tan \frac{x}{2}$

evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x - \cos x} dx$ 3

5) (a) Given $I_n = \int \sin^n x dx$, show that the expression for I_n in terms of I_{n-2} is given by $I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$ 4

Conics Section (Start a new page)

1) Find the distance between the foci of the ellipse $2x^2 + 3y^2 = 6$ 2

2) Consider the Cartesian Equation $4x^2 - 9y^2 = 36$. Find
 (a) the eccentricity 1
 (b) the foci coordinates 1
 (c) the equations of the directrices 1

3) Consider the hyperbola H with equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$
 (a) Show that the hyperbola H is satisfied by the parameters
 $x = 4 \sec \theta$ and $y = 3 \tan \theta$ 1

(b) Show that the equation of the tangent to the hyperbola H at the point $(4 \sec \theta, 3 \tan \theta)$ is $\frac{x \sec \theta}{4} - \frac{y \tan \theta}{3} = 1$. 3

- 4) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$.

Tangents to the rectangular hyperbola at P and Q intersect at $R(X, Y)$.

- (a) Show that the tangent to the rectangular hyperbola at $\left(ct, \frac{c}{t}\right)$ is
given by $x + t^2y = 2ct$

2

- (b) Show that $X = \frac{2cpq}{p+q}$, $Y = \frac{2c}{p+q}$

3

- (c) If P and Q are variable points on the rectangular hyperbola which move so
that $p^2 + q^2 = 2$, show that the equation of the locus of R is $y^2 + xy = 2c^2$

4

END OF EXAMINATION

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Extension two task 3 2015 (Solutions)

Multiple Choice

$$\begin{aligned} 1/ \quad u &= \ln x & \text{when } x &= e^4 \\ \frac{du}{dx} &= \frac{1}{x} & u &= \ln e^4 \\ x du &= dx & u &= 4 \quad (b) \\ & & \text{when } x &= e^3 \\ & & u &= 3 \quad (a) \end{aligned}$$

(B)

$$\begin{aligned} 2/ \quad \left[\ln(1+e^x) \right]_0^1 &= \ln(1+e) - \ln 2 \\ &= \ln \left(\frac{1+e}{2} \right) \end{aligned}$$

(C)

$$\begin{aligned} 3/ \quad \text{Asymptotes are } y &= \pm \frac{bx}{a} \\ &= \pm 2x \end{aligned}$$

(D)

$$4/ \quad \text{let } A=1, C=2, F=2$$

$$x^2 + 2y^2 = 2$$

$$\frac{x^2}{2} + y^2 = 1$$

(A)

Integration Section

$$\begin{aligned} 1/ \quad \int x(9-16x^2)^{-\frac{1}{2}} dx &= -\frac{1}{32} \int -32x(9-16x^2)^{-\frac{1}{2}} dx \\ &= -\frac{1}{32} \times (9-16x^2)^{\frac{1}{2}} \times \frac{2}{1} + C \\ &= -\frac{1}{16} \sqrt{9-16x^2} + C \end{aligned}$$

$$\begin{aligned} 2/ \quad x+1 \over x^2+0x+0 &= \frac{x-1}{x^2+x} \\ &= \frac{-x+0}{-x-1} \\ &= \frac{-x-1}{1} \end{aligned}$$

$$\int \frac{x^2}{x+1} dx = \int x-1 + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} - x + \ln|x+1| + C$$

$$\begin{aligned} 3/ \quad u &= x & \frac{dv}{dx} &= e^x \\ \frac{du}{dx} &= 1 & v &= e^x \\ \int_0^{\ln 3} x e^x dx &= \left[x e^x \right]_0^{\ln 3} - \int_0^{\ln 3} e^x dx \\ &= 3 \ln 3 - [3^0 - 1] \\ &= 3 \ln 3 - 2 \end{aligned}$$

$$4/a) \frac{2}{(x+1)(x^2+1)} = \frac{A(x^2+1)}{(x+1)(x^2+1)} + \frac{(Bx+C)(x+1)}{(x+1)(x^2+1)}$$

$$\therefore 2 \equiv A(x^2+1) + (Bx+C)(x+1)$$

Let $x = -1$	Let $x = 0$	Let $x = 1$
$2 = 2A$	$2 = 1 + C$	$2 = 2 + 2B + 2$
$1 = A$	$C = 1$	$-2 = 2B$
		$B = -1$

$$(b) \int_0^1 \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} dx$$

$$= \left[\ln|x+1| \right]_0^1 - \frac{1}{2} \left[\ln|x^2+1| \right]_0^1 + \left[\tan^{-1}x \right]_0^1$$

$$= \ln 2 - 0 - \left[\frac{1}{2} \ln 2 - 0 \right] + \left[\frac{\pi}{4} - 0 \right]$$

$$= \ln 2 - \frac{1}{2} \ln 2 + \frac{\pi}{4}$$

$$= \frac{1}{2} \ln 2 + \frac{\pi}{4}$$

$$(c) \quad t = \tan \frac{x}{2} \quad \frac{dx}{1+t^2} \quad \text{when } x=0, t=0 \\ x=\pi, t=1$$

$$\therefore = \int_0^1 \frac{2t}{1+t^2} \times \frac{2 dt}{1+t^2} = \int_0^1 \frac{2t}{1+t^2} dt = \left[\ln|1+t^2| \right]_0^1$$

$$= \int_0^1 \frac{4t}{(1+t^2)^2} dt$$

$$= \int_0^1 \frac{4t}{(1+t^2)} \times \frac{1}{2t^2+2t} dt$$

$$= \int_0^1 \frac{4t}{(1+t^2)} \times \frac{1}{2t(t+1)} dt$$

$$= \int_0^1 \frac{2}{(1+t^2)(t+1)} dt$$

$$= \frac{\pi}{4} + \frac{1}{2} \ln 2 \quad (\text{From (b)})$$

Conics Section

$$1/ \frac{x^2}{3} + \frac{y^2}{2} = 1$$

$$2 = 3(1 - e^2)$$

$$2 = 3 - 3e^2$$

$$3e^2 = 1$$

$$e = \frac{1}{\sqrt{3}}$$

$\therefore S$ is $(1, 0)$

S' is $(-1, 0)$

$\therefore$ Distance between foci is 2 units

$$2/ \frac{x^2}{9} - \frac{y^2}{4} = 1 \quad (a) \frac{4}{9} = e^2 - 1 \quad (b) \text{ foci are } (\pm ae, 0)$$

$$\frac{13}{9} = e^2$$

$$= (\pm \sqrt{13}, 0)$$

$$e = \frac{\sqrt{13}}{3}$$

$$(c) x = \pm \frac{9}{\sqrt{13}}$$

$$3/(a) \frac{16 \sec^2 \theta}{16} - \frac{9 \tan^2 \theta}{9} = \text{LHS}$$

$$1 + \tan^2 \theta - \tan^2 \theta = \text{LHS}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$(b) \frac{2x}{16} - \frac{dy}{dx} \frac{2y}{9} = 0$$

$$\frac{2x}{16} \times \frac{9}{2y} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{9x}{16y}$$

$$\text{at } x = 4 \sec \theta \quad y = 3 \tan \theta$$

$$m_T = \frac{9 \times 4 \sec \theta}{16 \times 3 \tan \theta}$$

$$= \frac{3 \sec \theta}{4 \tan \theta}$$

$$5) I_n = \int \sin x \sin^{n-1} x \, dx$$

$$u = \sin^{n-1} x$$

$$\frac{du}{dx} = (n-1) \sin^{n-2} x \cos x$$

$$\frac{dv}{dx} = \sin x$$

$$V = -\cos x$$

$$I_n = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

$$= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x - \sin^n x \, dx$$

$$I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + n I_n - I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$$

$$n I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$$

$$\therefore I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{(n-1)}{n} I_{n-2}$$

$$y - 3\tan\theta = \frac{3\sec\theta}{4\tan\theta} (x - 4\sec\theta)$$

$$4y\tan\theta - 12\tan^2\theta = 3x\sec\theta - 12\sec^2\theta$$

$$12\sec^2\theta - 12\tan^2\theta = 3x\sec\theta - 4y\tan\theta$$

$$12(1 + \tan^2\theta - \tan^2\theta) = 3x\sec\theta - 4y\tan\theta$$

$$12 = 3x\sec\theta - 4y\tan\theta$$

$$1 = \frac{x\sec\theta}{4} - \frac{y\tan\theta}{3}$$

(as required)

$$4/ \quad y = cx^{-1}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2} \quad \text{at } x = ct$$

$$m_T = \frac{-c^2}{c^2 t^2}$$

$$= -\frac{1}{t^2}$$

$$y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$t^2 y - ct = -x + ct$$

$$t^2 y + x = 2ct \quad (\text{as required})$$

$$(b) \text{ Tangent at } P \quad p^2 y + x = 2cp$$

$$\text{Tangent at } Q \quad q^2 y + x = 2cq$$

$$\therefore 2cp - p^2 y = 2cq - q^2 y$$

$$2cp - 2cq = p^2 y - q^2 y$$

$$2c(p - q) = y(p - q)(p + q)$$

$$\frac{2c}{p + q} = y$$

$$\therefore y = \frac{2c}{p + q}$$

$$x = 2cp - \frac{p^2(2c)}{p + q}$$

$$x = \frac{2cp(p + q) - 2cp^2}{p + q}$$

$$x = \frac{2cp^2 + 2cpq - 2cp^2}{p + q} \therefore X = \frac{2cpq}{p + q}$$

$$(c) \quad p^2 + q^2 = 2$$

$$(p+q)^2 - 2pq = 2$$

$$\boxed{(p+q)^2 = 2 + 2pq} \quad (1)$$

$$\frac{x}{y} = \frac{2c pq}{(p+q)} \times \frac{(p+q)}{2c}$$

$$\boxed{\frac{x}{y} = pq} \quad (2)$$

$$\boxed{p+q = \frac{2c}{y}} \quad (3)$$

Sub (3) + (2) into (1)

$$\frac{4c^2}{y^2} = 2 + \frac{2x}{y}$$

$$4c^2 = 2y^2 + 2xy$$

$$\therefore y^2 + xy = 2c^2$$